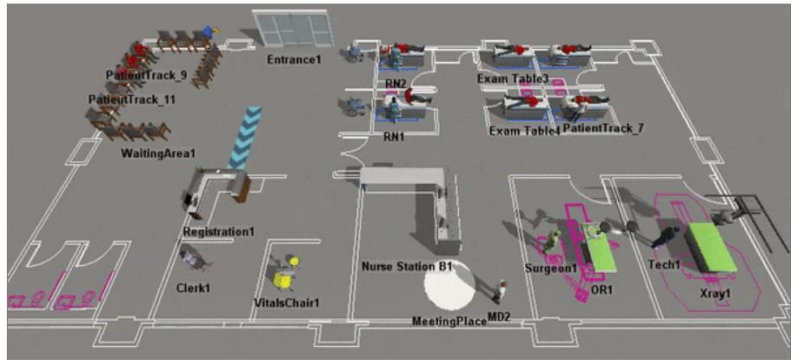
REPUBLIC POLYTECHNIC

Lesson 12
Healthcare Simulation of Clinic Operations (II)

E211 – Operations Planning II



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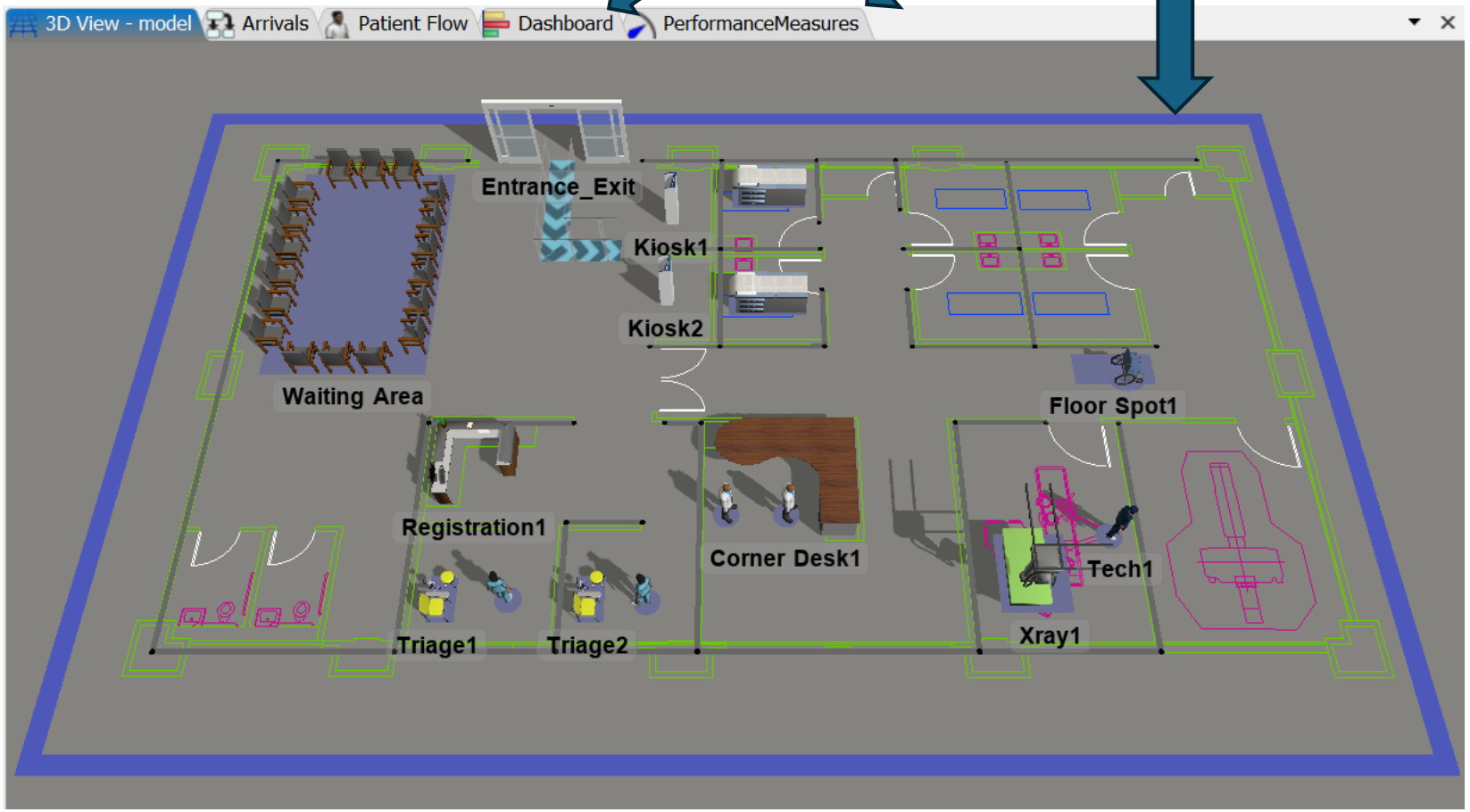
Learning Outcomes

At the end of the lesson, students should be able to:

- Describe the basic concept of Process Flow modelling technique in FlexSim Healthcare for a simple patient flow simulation.
- Build a Healthcare simulation model using 3D Healthcare objects and Patient Process Flow.
- Identify the 3D objects from FlexSim Healthcare library used to model a walk-in clinic.
- Identify the functions of the HC activities (HC Activity Sets, HC Basic, HC Sub Flows, HC Resources) from the patient flow library used to model the patient flow in a clinic.

Preview of Complete the Healthcare Simulation Model

To be added this lesson



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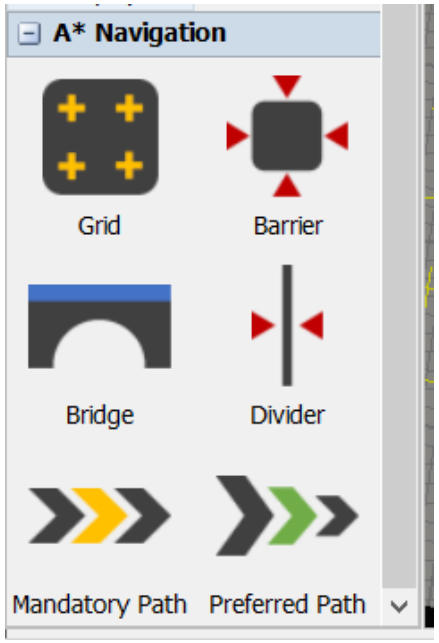
Part 3: A* Navigation & Dashboard

- Add A* Navigation and dividers to 3D model

- Create Dashboard to display these performance measures
    a) Milestone-Milestone Times
    b) Average Wait Time Charts for Staff & Location
    c) Average State Charts for Staff & Location
    d) Average Walking Distance
    e) Patient Census Chart

A* Navigation

What is A* Navigation?

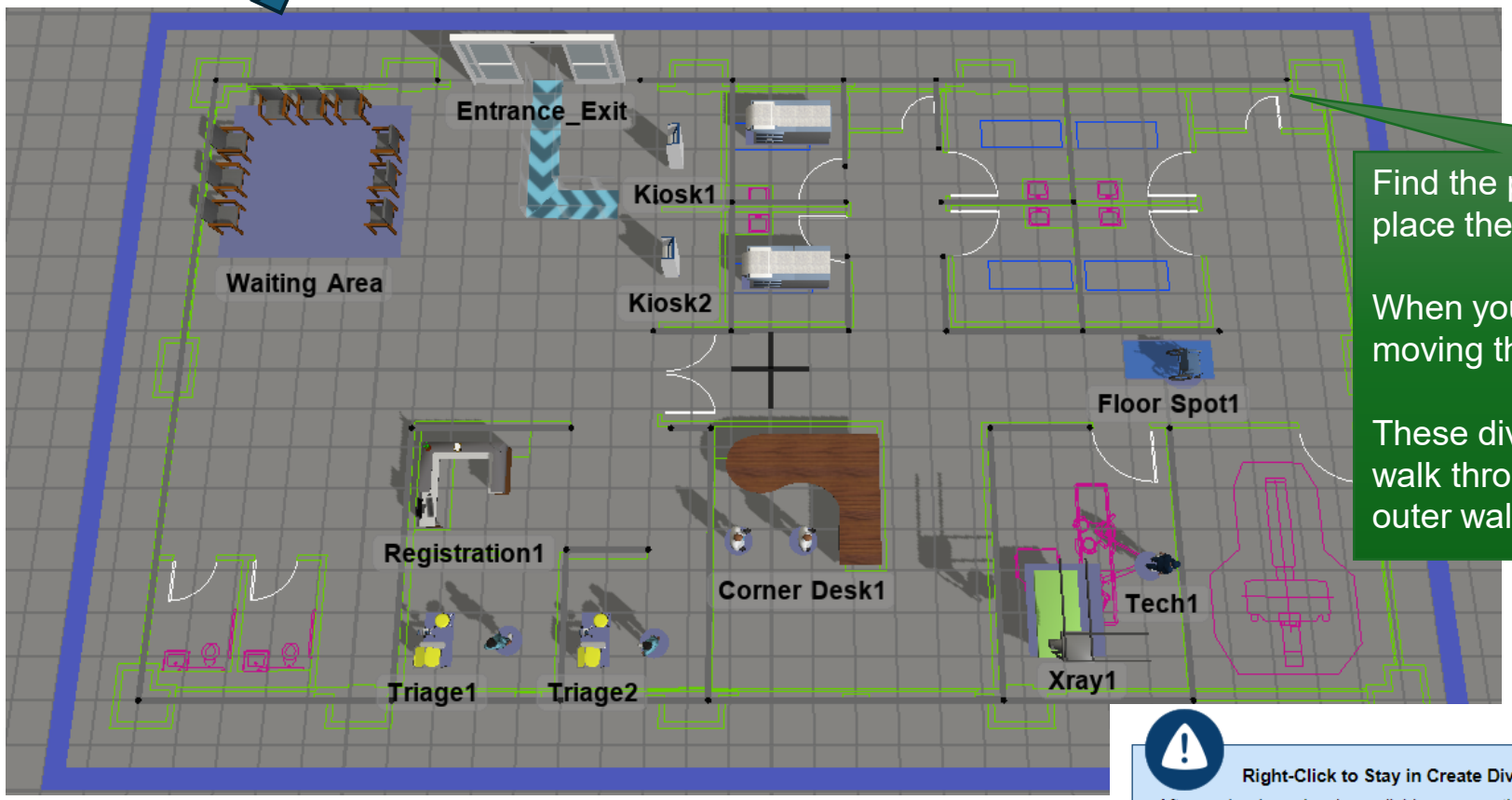


- A* Navigation is a tool that determines the shortest path for a person to walk from one object to another.
- It enables people to move along logical paths without walking through walls or into physical objects.
- Steps in using A* Navigation:

  a) Add Grid under A* Navigation
  b) Add Dividers
  c) Add 3D Objects to the A* Navigator

Add A* Navigation Grid & Dividers

First add a Grid. Drag to enlarge and encompass the area around the clinic.



Find the position in your model where you want to place the beginning of a divider.

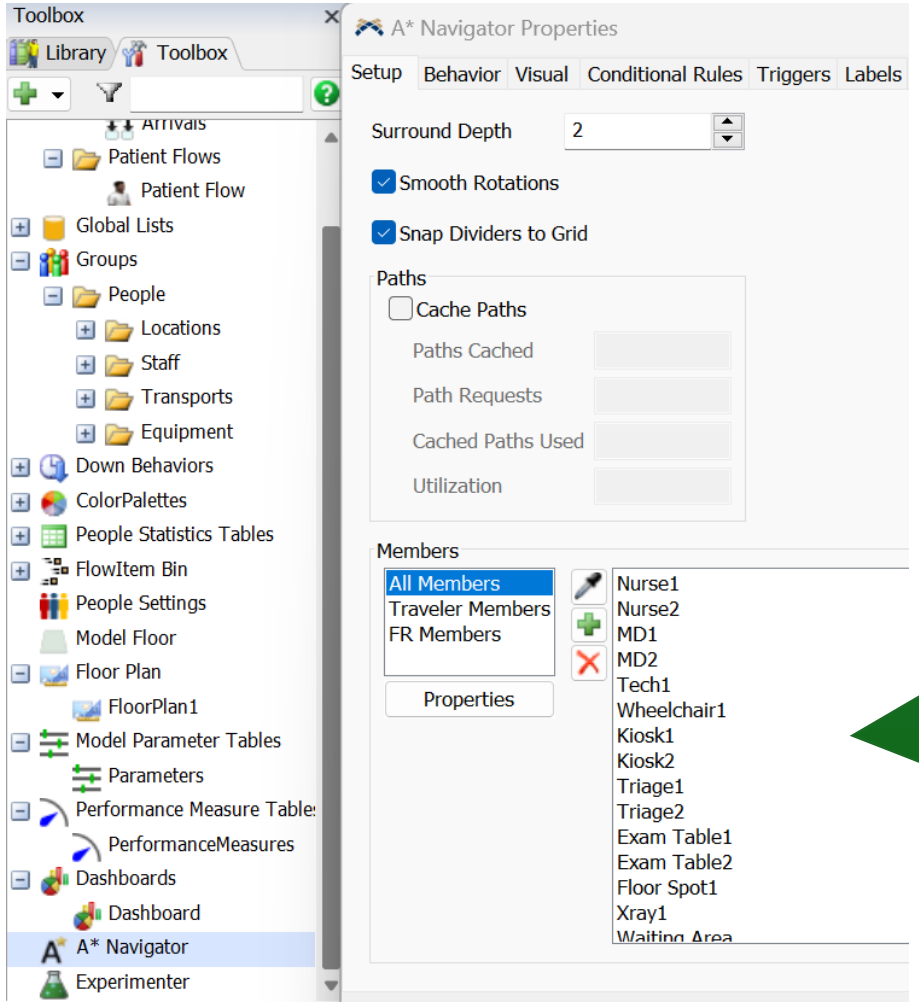
When you click that position in the model and start moving the cursor, it will create a divider.

These dividers will act as walls which people cannot walk through in the simulation model. Create the outer walls first, then the inner dividers.

Right-Click to Stay in Create Dividers Mode

After you're done drawing a divider or a continuous set of dividers, you can right-click to close off a divider but still remain in Create Dividers mode. You can use this method to create multiple dividers that aren't connected to each other.

Add Objects to A* Navigation



Access the A* Navigator Properties from Toolbox.

In the Setup tab, select All Members under Members window.

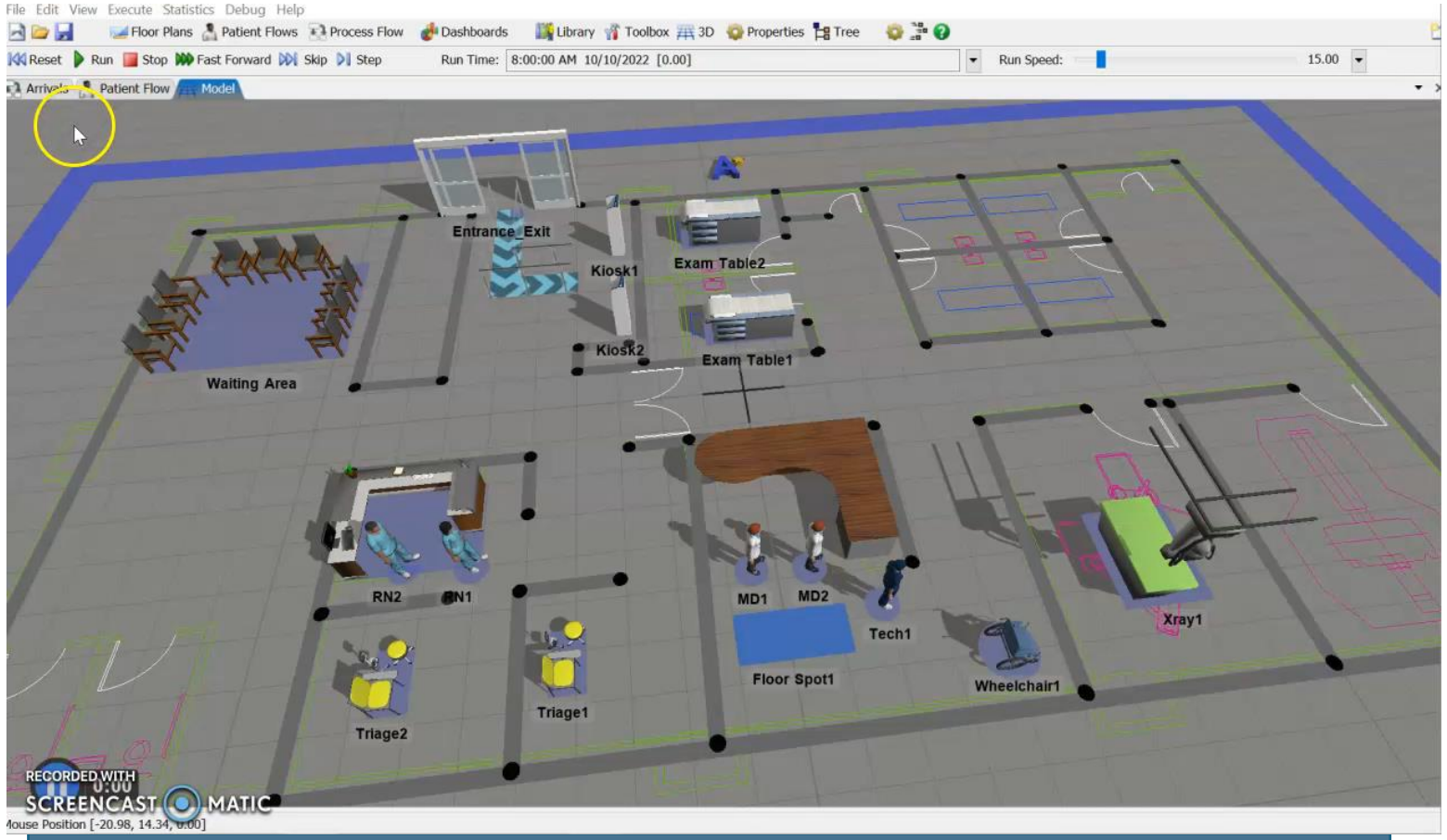
Click the Add button to open a menu of all the objects organized by their type.

Click the Locations, Prop, MultiLocations, Staff and Transport. Then click the Select button.

All the objects of the selected categories will now appear in the members list. They form part of the Navigator and patients will not walk through them.

Simulation Check

Run your simulation model and observe the movement of people.



Now when you run the model, the patients and staff are no longer walk through objects or walls. Notice that the nurses and doctors walks around the desks and the walls when traveling to the patient.

Dashboard

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Performance Measurements for the Healthcare Model

- Create the following dashboard charts for analyzing the performance of a patient flow model:

  a) Milestone-Milestone Times
  b) Average Wait Time Charts for Staff & Location
  c) Average State Charts for Staff & Location
  d) Average Walking Distance
  e) Patient Census Chart

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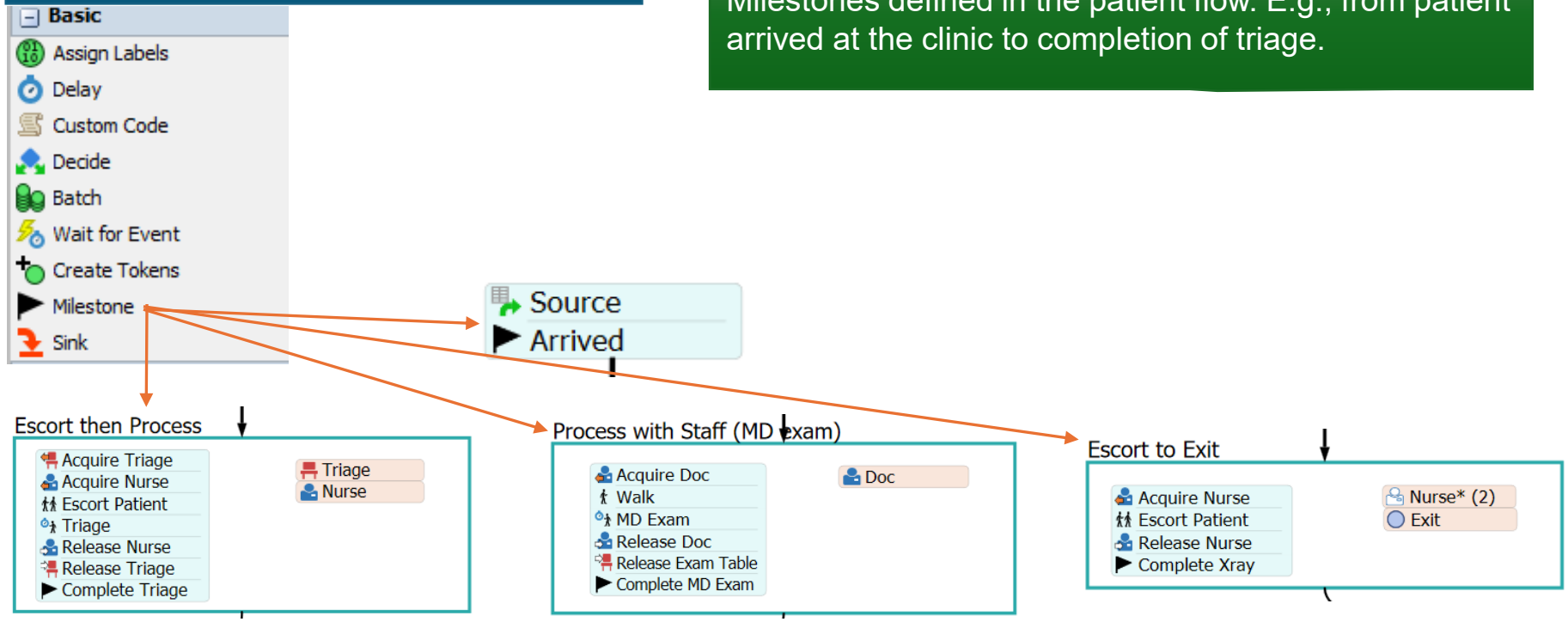
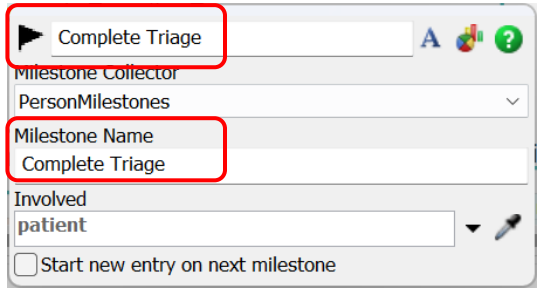
Milestone-to-Milestone Times

1. Under Basic, insert Milestones at the relevant parts of Patient Flow.

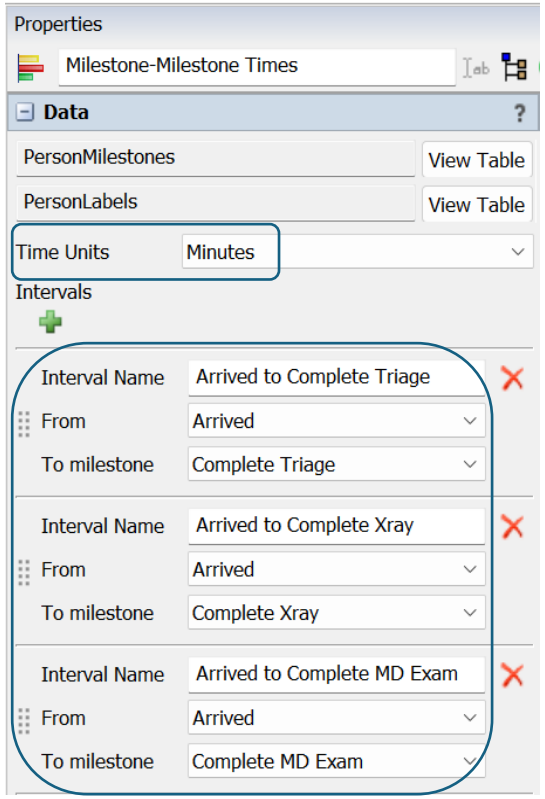
Milestone-to-Milestone time is the period between 2 Milestones defined in the patient flow. E.g., from patient arrived at the clinic to completion of triage.



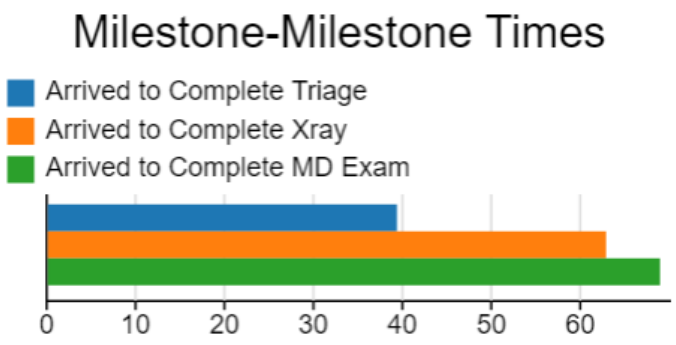
2. Change each Milestone Name and the activity name accordingly.



Add Dashboard & Milestone-to-Milestone Chart



1. Add a blank Dashboard.
2. Under the Library, select the Milestone-Milestone Times (Bar Chart) and add to the dashboard.
3. In the properties page, change the time units from Seconds to Minutes.
4. Create the 3 Milestone intervals as shown.

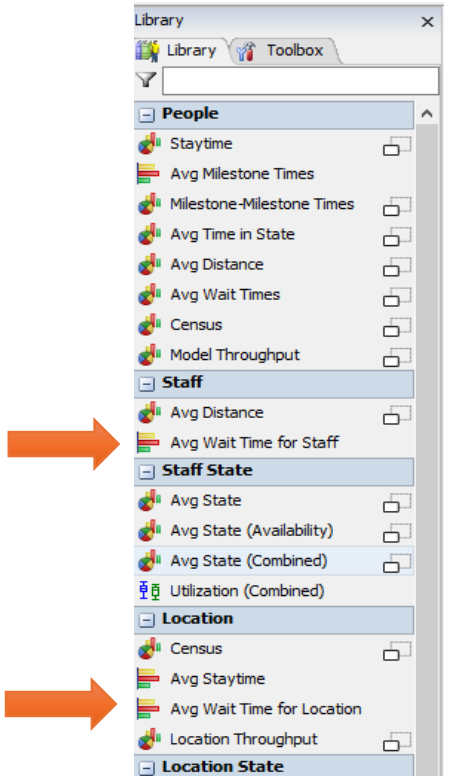


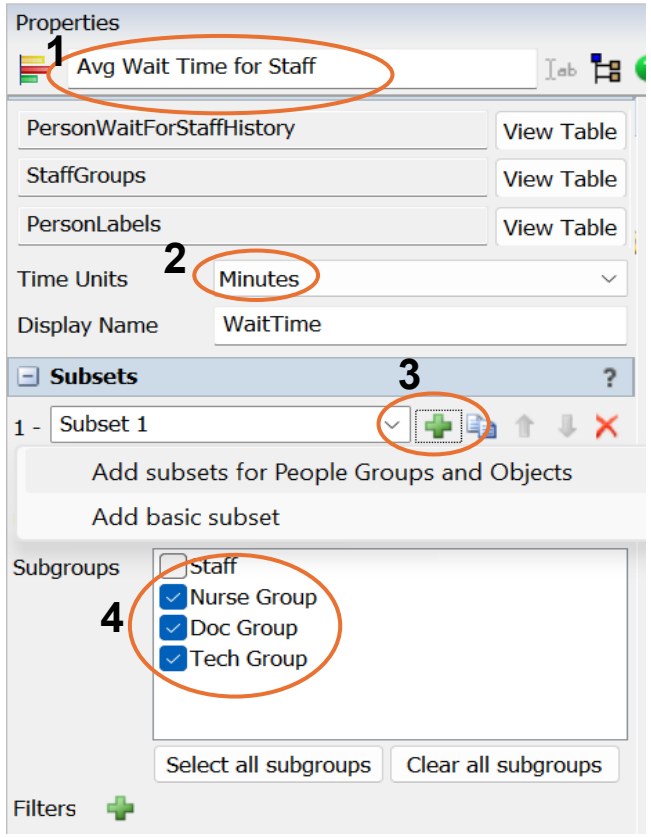
Video of Milestone-Milestone Bar chart creation

Milestone2milestone.mp4

Avg Wait Time for Staff and Location

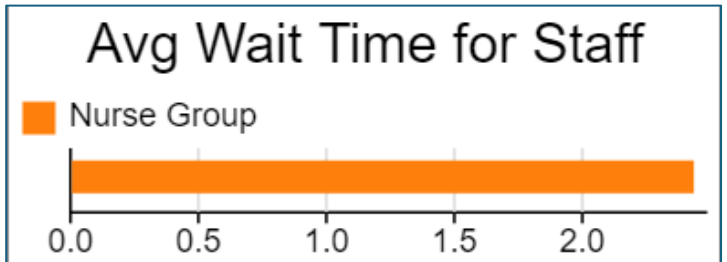
At the library menu, add Avg Wait time for staff (under **Staff**). Follow the settings shown below.



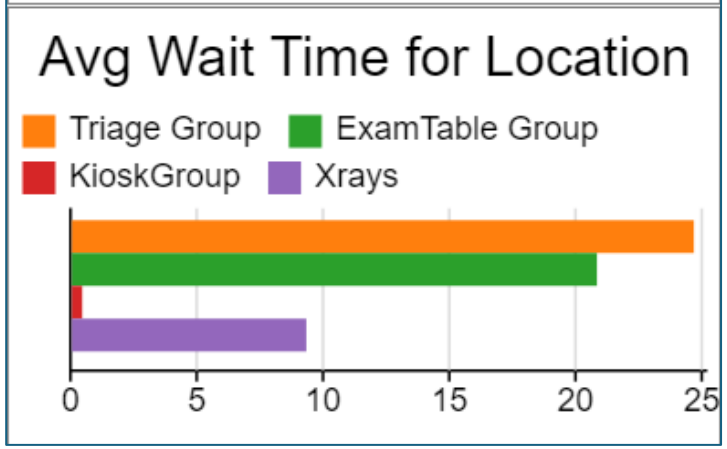


In addition, create a chart for Avg Wait time for Location (under **Location)** with the appropriate settings.

Avg Wait Time for Staff and Location



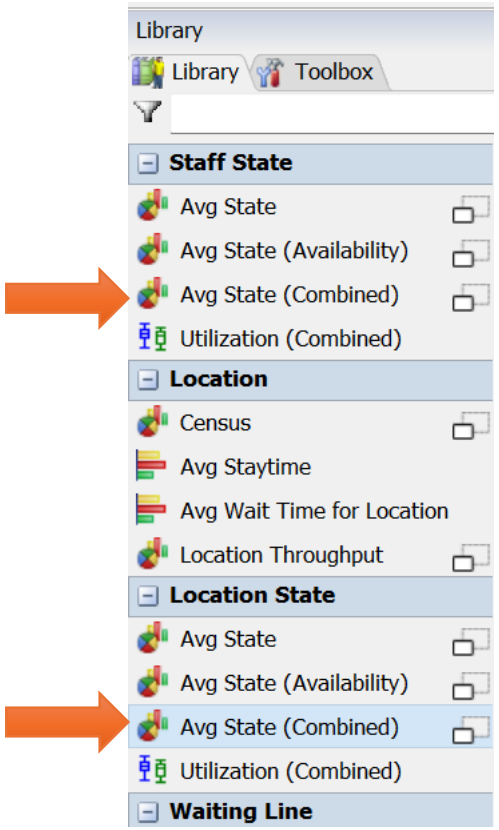
Why are Doc and Tech waiting times not displayed?

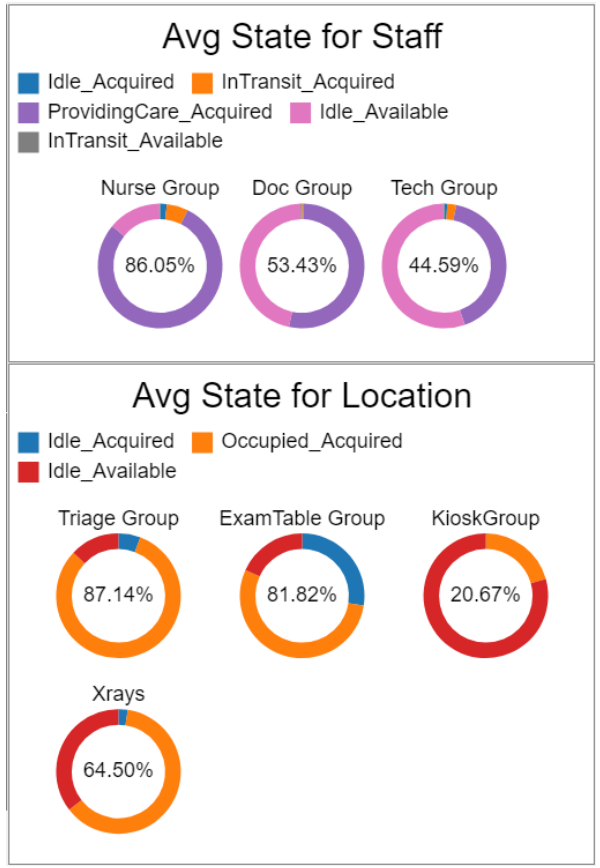


Avg State for Staff and Location

At the library menu, add two Avg State (Combined) charts, one under **Staff State**, and the other under **Location State.**

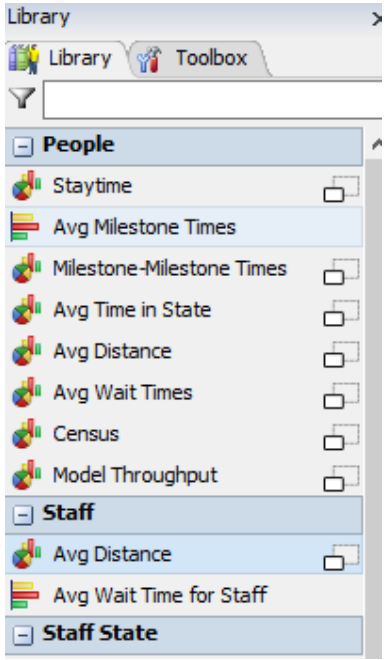
Select all the groups to compare under each chart's properties.

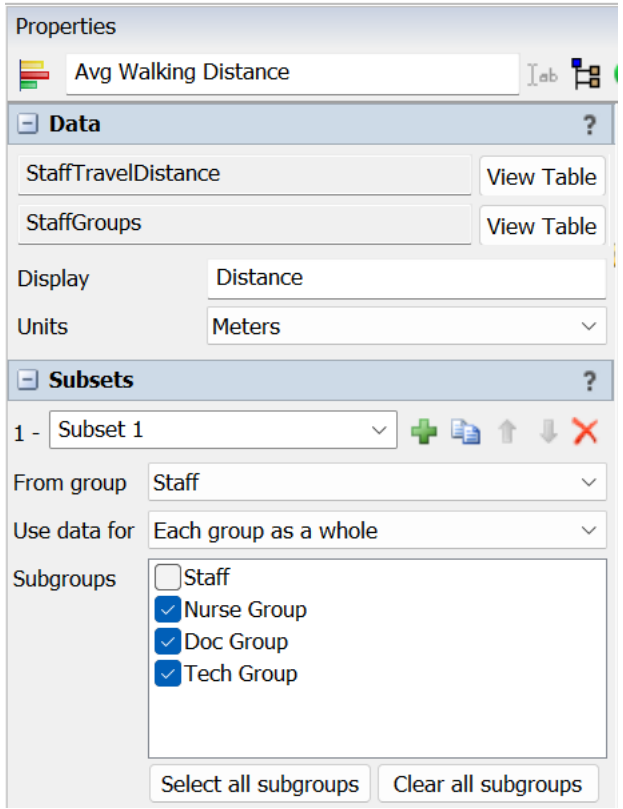


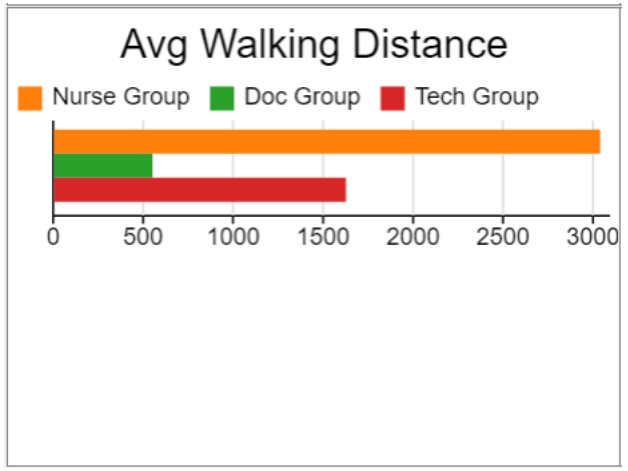


Avg Distance Chart

At the library menu, add Avg Distance (under **Staff**).

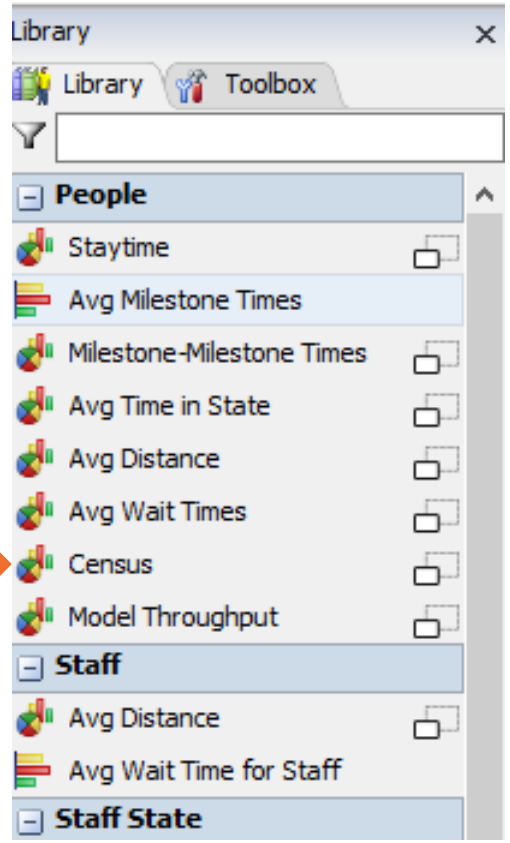


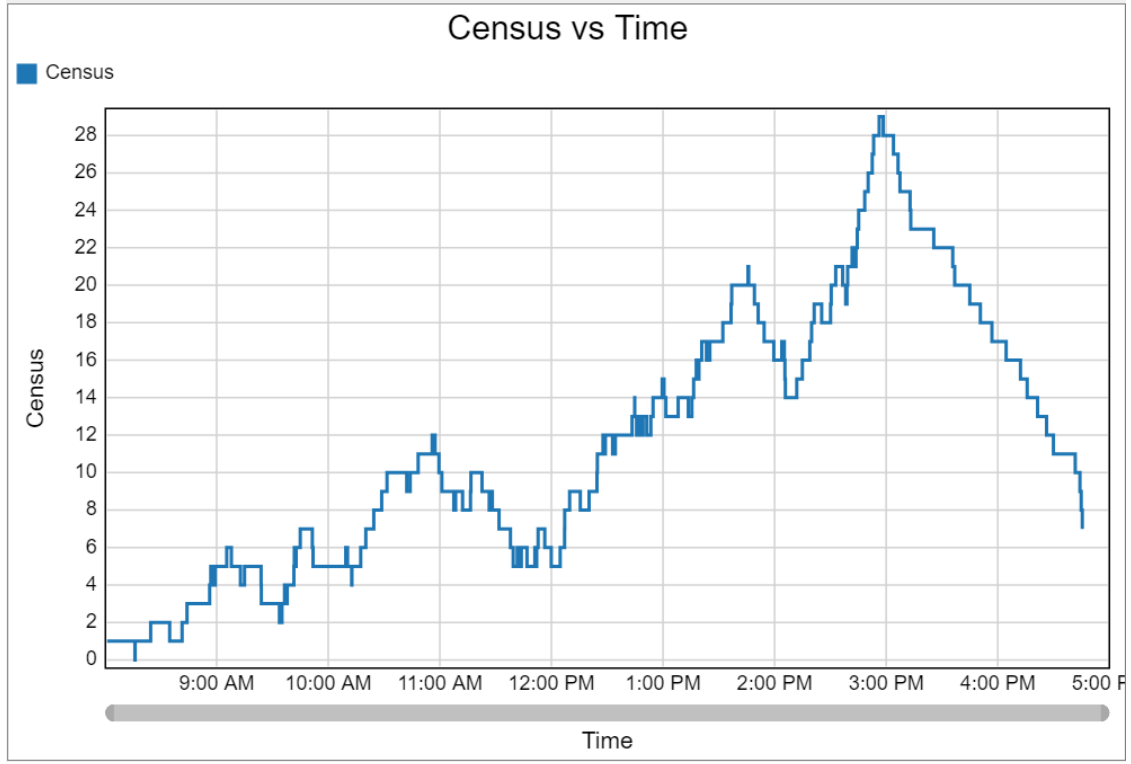




Patient Census

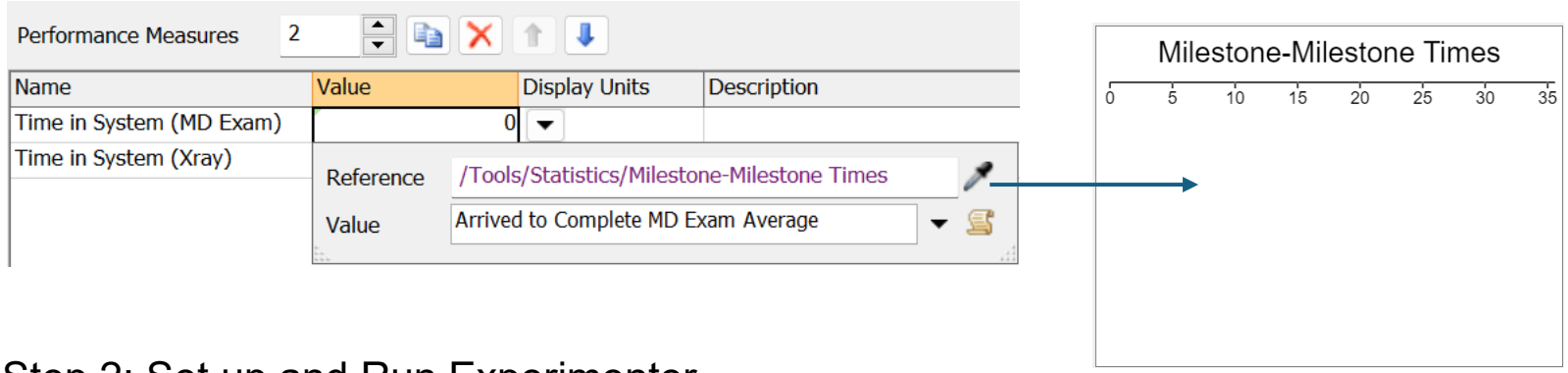
At the library menu, add a Census (under **People**) line chart. This chart shows how many patients are in the clinic at any given time. It is a useful chart to show the peak number of patients during the clinic's operating hours.



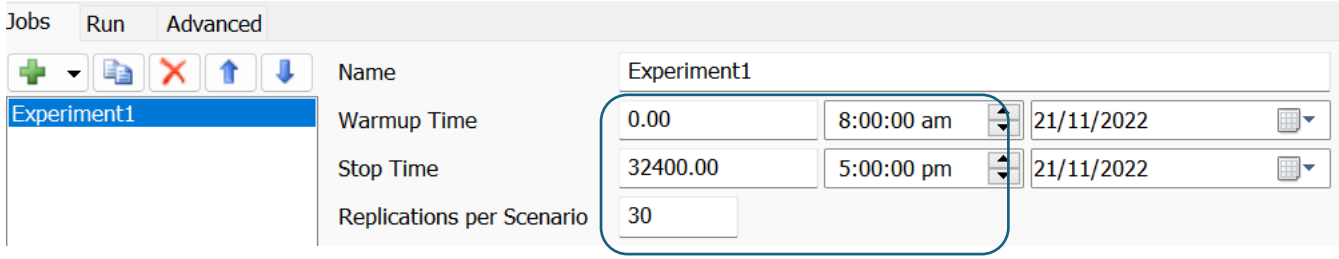


Set Up Performance Measures for Experimenter

Step 1: Set up two Performance Measures – Time in System (MD Exam path) and Time in System (Xray path). Use the Reference's sampler icon to select the Milestone-Milestone Times chart and select the respective milestone (average).



Step 2: Set up and Run Experimenter.



Experimenter Results

90% confidence interval for Time in System.

Time in System (MD Exam) | Data Summary | Mean Based on | 90% Confidence

Time in System (MD Exam)

	Mean (90% Confidence Interval)	Sample Std Dev	Min	Max
Current Scenario	82.75 ± 5.31	17.12	57.72	123.62

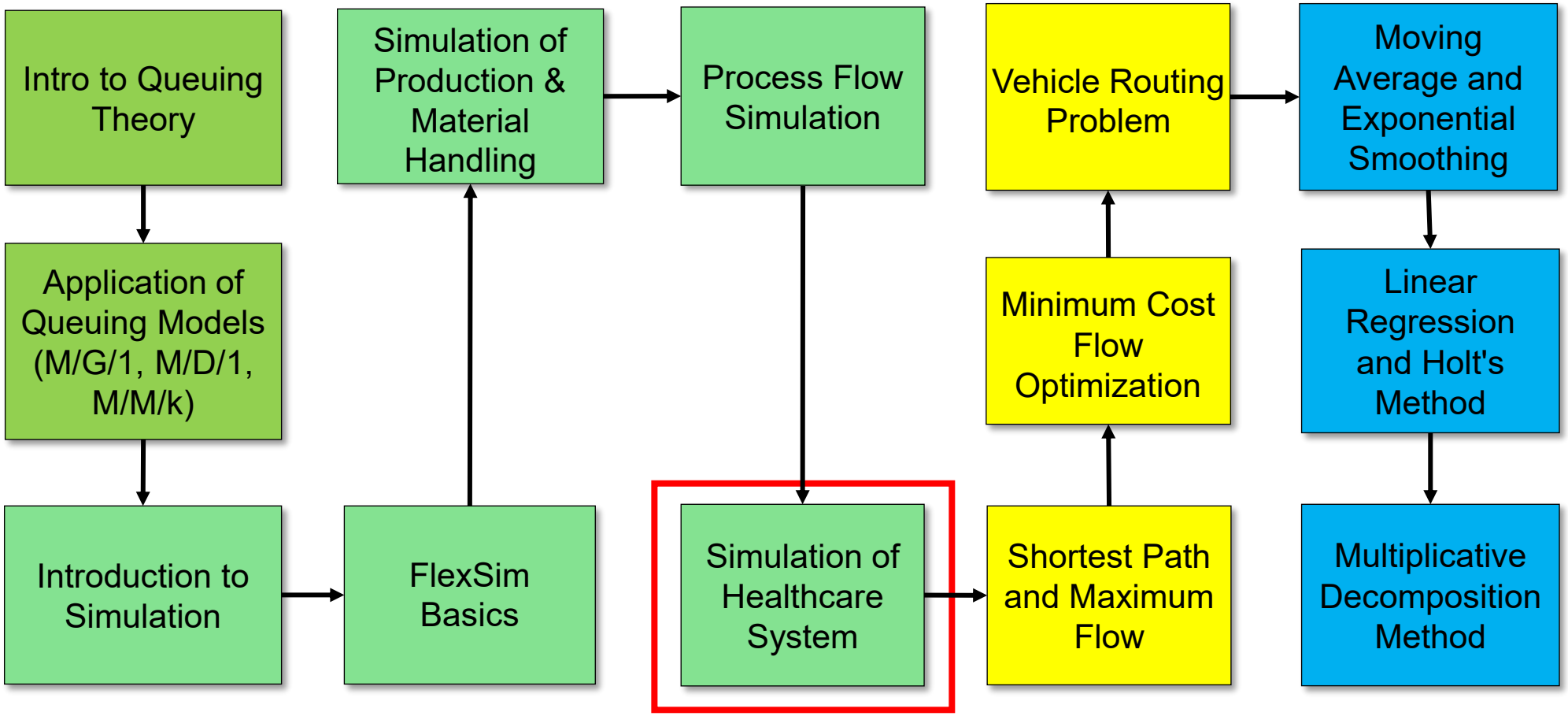
Time in System (Xray) | Data Summary | Mean Based on | 90% Confidence

Time in System (Xray)

	Mean (90% Confidence Interval)	Sample Std Dev	Min	Max
Current Scenario	72.64 ± 4.04	13.03	49.04	102.84

Experimentation

- Following patients' complaints about the long stay in the clinic, the clinic decides to set a service standard where the mean Time in System for patients should be 60 mins or less.

- Determine what changes are required to achieve the performance target set by the clinic. All patients should complete their treatment by 5pm.

E211 Progression